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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
B.Sc Degree Examination – APRIL 2021

PH612S – LASER AND FIBER OPTIC COMMUNICATION

Time : 3 hrs **Max Marks :75**

SECTION – A (10X2=20)

Answer **ALL** Questions

1. State the Einstein Coefficients.
2. Explain – Line shape function.
3. What are Gas lasers?
4. Define Argon ion.
5. What is surface Hardening?
6. Explain Holography.
7. Define acceptance angle.
8. Distinguish between step index and graded index fibers.
9. What are Analog and digital modulation methods.
10. Write a note on APD photodetectors.

SECTION – B (5X5=25)

Answer **ALL** Questions

11. a) Explain the condition for light amplification.

(or)

b) Explain three level and four level systems.

12. a) Explain Heterojunction lasers.

(or)

b) Explain Eximer Laser.

13. a) Write the Application of laser in industry.

(or)

b) Explain the laser as diagnostic and therapeutic tool.

14. a) Write the basic characteristics of optical fiber.

(or)

b) Explain the single mode and multi mode fibers.

15. a) Explain the Source and detectors for fiber optic communication.

(or)

b) Discuss the principle of Laser and LED.

SECTION- C (3X10=30)

Answer any **THREE** Questions

16. Describe Basic Principle of Laser and explain Einstein Coefficients.

17. Explain the Q switching and mode locking.

18. Explain the cutting welding, Drilling & surface Hardening. Also discuss the Medical applications of laser.

19. Explain the theory of mode formation and the classification of fibers.

20. What is the principle of optical detection? Explain the pin and APD photo detectors.

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